I recently read a blog asking the question, "Do decks matter in Clash Royale, or is it skill and luck?" While this is a good question to ask, I noticed that the author attempts to answer it using a classifier and classifier metrics.

I believe the metric chosen answers a different question than the one asked. Accuracy is a *whether* metric and that It tells you a signal exists not how impactful it is.

Beating a 50% baseline beats a pretty low bar, and clearing it says little about whether the signal is deck strength, the player holding the deck, or data leakage.

Take this for example. Suppose I threw out every card name in the dataset and kept exactly one feature per player: how many evolution cards they have in their decl.

*insert image of dataset*

A model trained on this would certainly not learn anything about card synergies, combinations, or counters. The model wouldn't be able to tell a Hog cycle from a Golem beatdown, but it would almost certainly beat 50% with $p < 0.001$, because owning evolutions tracks how much time and money someone has poured into their account, and that tracks skill. Same metric, same significance, and the conclusion "decks matter" would be flatly wrong. (I'd have measured wallet size and called it card strength.)

---